

Tab. 2: Models selected by AIC for each vital rate. x is the size and a the age. The random effects are noted $year$, pop , $year : pop$ (for population nested in year effect), and ε for residual variance.

Survival probability for seedling

$$\text{logit}(s_1(x)) = \begin{cases} year_0 - 1.66(x - 0.5) + (3.99 + year_1)(x - 0.5)^2 & \text{if } 0.5 \leq x \leq 1 \text{ cm} \\ year_0 + 0.166 + (2.33 + year_1)(x - 1) - 2.3(x - 1)^2 & \text{if } 1 \leq x \leq 1.5 \text{ cm} \\ year_0 + 0.756 + (0.0292 + year_1)(x - 1.5) + 0.0502(x - 1.5)^2 & \text{if } 1.5 \leq x \leq 15 \text{ cm} \end{cases}$$

$$Var(year_0) = 1.582; Var(year_1) = 0.02833; Cov(year_0, year_1) = -0.04404271$$

Survival probability for rosettes

$$\text{logit}(s_2(x)) = pop_0 + year_0 + \text{spline}_1(x) + \text{spline}_2(a)$$

$$\text{spline}_1(x) = \begin{cases} 1.01(x - 0.5) - 0.0958(x - 0.5)^2 & \text{if } x \leq 2 \\ 2.52 + 0.246(x - 4.5) - 0.0117(x - 4.5)^2 & \text{if } x \geq 4.5 \end{cases}$$

$$\text{spline}_2(a) = \begin{cases} (0.713 + year_1)(a - 2) - 0.478(a - 2)^2 + 0.0709(a - 2)^3 & \text{if } a \leq 6.5 \\ (0.723 + year_1)(a - 6.5) + 0.48(a - 6.5)^2 - 0.952(a - 6.5)^3 & \text{if } a \geq 6.5 \end{cases}$$

$$Var(pop_0) = 0.076; Var(year_0) = 1.443; Var(year_1) = 0.011; Cov(year_0, year_1) = -0.014$$

Flowering probability

$$\text{logit}(f(x, a)) = -10.21 + pop_0 + 226.93x - 76.57x^2 + 39.27x^3 + (133.48 + pop_1)a - 54.57a^2$$

$$Var(pop_0) = 2.252; Var(pop_1) = 0.09228; Cov(pop_0, pop_1) = -0.2041362$$

Plant growth

$$g(y^\alpha, x, a) = \frac{1}{\sigma(x, a)\sqrt{2\pi}} \exp\left(-\frac{(y^\alpha - \mu(x, a))^2}{2\sigma(x, a)^2}\right)$$

$$\mu(x, a) = pop_0 + year_0 + poly(x^\alpha) + \text{spline}(a)$$

$$poly(x) = 2.204 + (27.491 + year_1)x^\alpha + 0.352x^{2\alpha} - 3.111x^{3\alpha}$$

$$\text{spline}(a) = \begin{cases} (-0.161 + year_2)(a - 1) + 0.023(a - 1)^2 & \text{if } a \leq 6.5 \\ -0.192 + (0.091 + year_2)(a - 6.5) - 0.106(a - 6.5)^2 & \text{if } a \geq 6.5 \end{cases}$$

$$\sigma(x, a) \sim \text{Gamma}(\log(-1.832 + pop'_0 + year'_0 - 0.053 \log(x) - 0.054 \log(a) + \varepsilon))$$

$$Var(pop_0) = 0.00518; Var(year_0) = 0.014; Var(year_2) = 4.14e^{-4}; Cov(year_0, year_2) = -2.56e^{-4};$$

$$Var(pop'_0) = 0.03; Var(year'_0) = 0.0197;$$

$$\alpha = 0.4343354$$

Fecundity (number of capitulas)

$$\log(c(x, a)) = 2.32734 + year_0 + 0.06752x + year_1a + \varepsilon$$

$$Var(year_0) = 0.2127; Var(year_1) = 0.0238; Cov(year_0, year_1) = -0.00506226; Var(\varepsilon) = 0.282$$

Seedling size distribution

$$\omega(y) \sim \text{Gamma}(\log(0.2873 + pop_0 + year_0 + (pop : year_0) + \varepsilon))$$

$$Var(year_0) = 0.0249; Var(pop_0) = 0.0248; Var(pop : year_0) = 0.0781; Var(\varepsilon) = 0.0213$$

Establishment rate

$$\#Seedlings \sim \text{Poisson}(\log(\lambda(\#Capitula)))$$

$$\lambda(\#Capitula) = -1.643 + \#Capitula + (pop : year_0)$$

$$Var(pop : year_0) = 0.7155$$
